with semiconductor atoms while passing through the depletion region. This leads to the breaking up of covalent bonds and this results in the generation of electron - hole pairs.

The newly generated charge carriers are also accelerated by the reverse voltage resulting in more collisions and further production of charge carriers. This cumulative process leads to an avalanche (uncontrollably large number) of charge carriers across the junction. This causes diode current to rise abruptly and breakdown takes place. This breakdown is called avalanche breakdown.

ii) Zener breakdown

Heavily doped *p-n* junctions have narrow depletion layers whose width is of the order of $<10^{-6}$ m. When reverse voltage across this junction is increased to the breakdown limit, a very strong electric field of strength $3 \times 10^{7}\,\mathrm{V\,m^{-1}}$ is set up across the narrow layer. This electric field is strong enough to break or rupture the covalent bonds in the lattice and thereby generating electron-hole pairs. This effect is called **Zener effect**.

Even a small further increase in reverse voltage produces a large number of charge carriers which move across the junction through the thin depletion region. This process gives rise to a large amount reverse current or breakdown current and this breakdown is called Zener breakdown.

Note

In Avalanche breakdown, the minority charge carriers gain sufficient energy from excessive reverse bias voltage to break covalent bond in order to produce new charge carriers. But Zener breakdown occurs due to the direct rupture of covalent bonds because of the existence of the strong electric field. Since depletion region is thin, Zener breakdown occurs usually at lesser reverse bias voltage compared to Avalanche breakdown voltage.

10.3.6 Zener diode

Zener diode is a heavily doped silicon diode used in reverse biased condition and is named after its inventor Clarence Melvin Zener. It is specially designed to be operated in the breakdown region. The doping level of the silicon diode can be varied to have a wide range of breakdown voltages from 2 V to over 1000 V.

As explained in the previous section, Zener breakdown occurs due to the breaking up of covalent bonds by the strong electric field set up in the depletion region by the reverse voltage. It produces an extremely large number of electrons and holes which constitute the reverse saturation current. The current is limited by both external resistance and power dissipation of the diode. A Zener diode is shown in Figure 10.19(a) and its circuit symbol is given in Figure 10.19(b).

It looks like an ordinary *p-n* junction diode except that *n*-side lead resembles the shape of the letter 'z'. The arrow head points the direction of conventional current. In Figure 10.19(a), black ring indicates the *n*-side lead.



Figure 10.19 Zener diode
(a) Commercial picture (b) Circuit symbol

V-I Characteristics of Zener diode

The circuit to study the forward and reverse characteristics of a Zener diode is shown in Figure 10.20(a) and Figure 10.20 (b). The V-I characteristics of a Zener diode is shown in Figure 10.20(c). The forward characteristic of a Zener diode is similar to that of an ordinary *p-n* junction diode. It starts conducting approximately around 0.7 V. However, the reverse characteristics is highly significant in Zener diode. The increase in reverse voltage

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